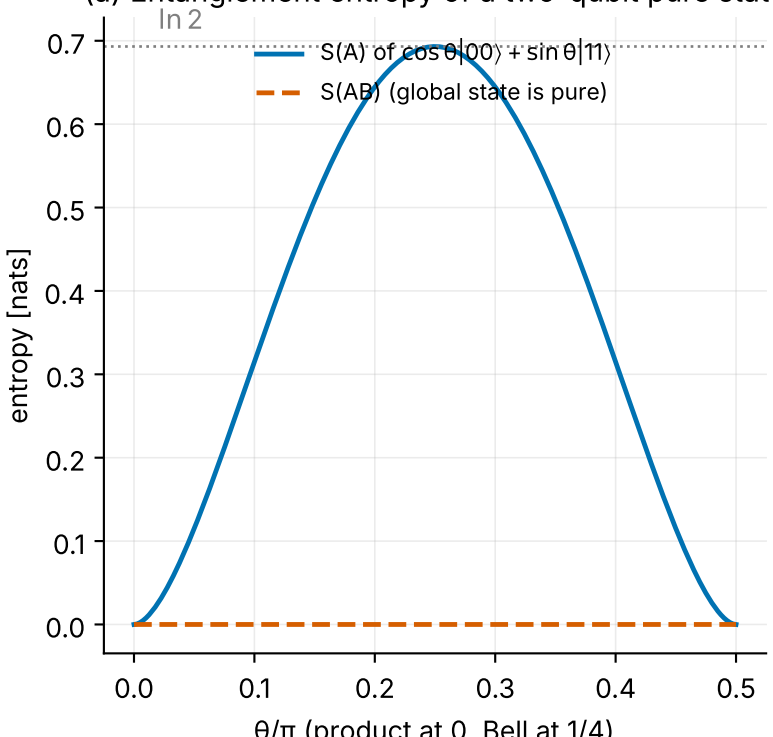
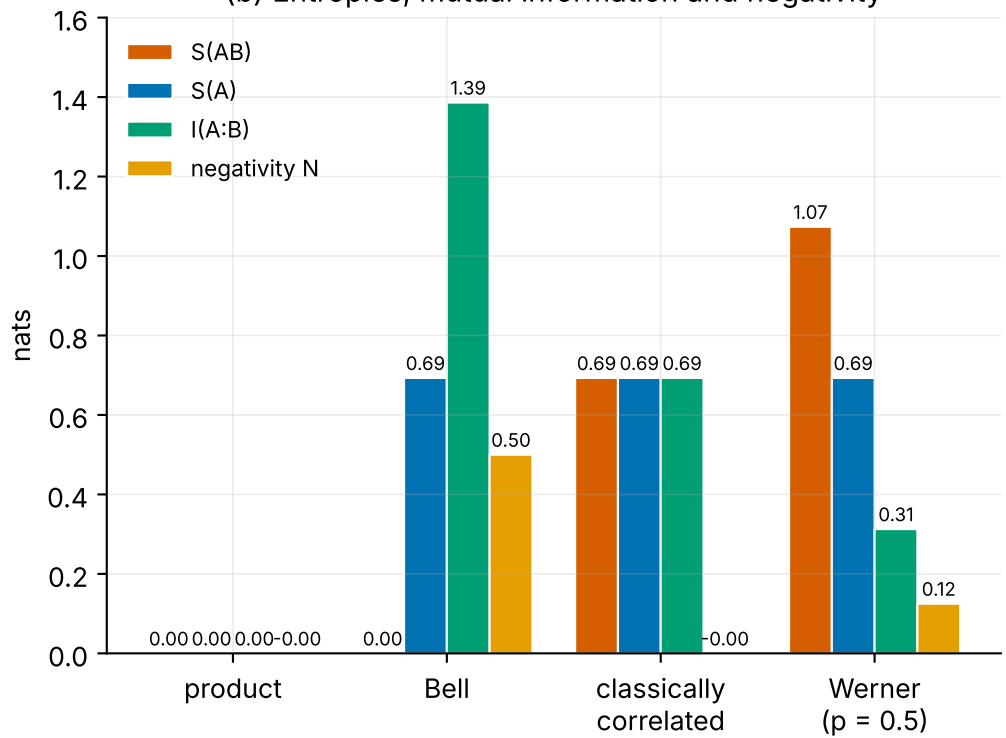

(a) Entanglement entropy of a two-qubit pure state
ln 2
S(A) of cos θ|00⟩ + sin θ|11⟩
S(AB) (global state is pure)
entropy [nats]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.0
0.1
0.2
0.3
0.4
0.5
θ/π (product at 0, Bell at 1/4)


(b) Entropies, mutual information and negativity
S(AB)
S(A)
I(A:B)
negativity N
nats
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
0.00 0.00 0.00 -0.00
0.00
0.69
1.39
0.50
0.69 0.69 0.69
-0.00
1.07
0.69
0.31
0.12
product
Bell
classically correlated
Werner (p = 0.5)